MP4

DOM 027

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Pranjal

MP4: Logistic Regression Analysis of Women's Labor Force Participation

1. Introduction

This project aims to model and understand the factors that influence women's labor force participation using logistic regression. The objective is to follow a structured statistical workflow—beginning with data cleaning and preprocessing, followed by visualization, hypothesis-driven model building, and diagnostic evaluation—to produce a final, defensible model aligned with the MP4 specifications.**
This study investigates the determinants of women's labor force participation using micro-level survey data. The dependent variable (inlf) indicates whether a married woman is engaged in the labor force (1 = Yes, 0 = No). Guided by human capital theory and household labor supply models, we examine how demographic characteristics, education, labor market experience, household income, and childcare responsibilities influence the probability of participation.

The goal of this MP4 is to estimate a logistic regression model, evaluate key hypotheses, test nonlinearities and interactions, perform model diagnostics, and present a final parsimonious model supported by statistical evidence and economic rationale.

2. Data Pre-Processing

Before analysis, several preprocessing steps were performed to ensure data quality and suitability for logistic regression:

The excel that was given to us was almost fit for processing however there were a lot of values that were not filled in properly such as the column containing the wage data, this was later confirmed by Jaideep sir that the dots are supposed to be zero vales

D	E	F	G	H	I	J
1	34	11	.	0	1804	
1	31	8	.	0	2805	
0	56	12	.	0	2160	
1	42	16	.	0	2052	
2	32	12	.	0	2250	
3	35	12	.	0	1120	
1	30	12	.	0	2450	
0	51	10	.	0	3432	
3	47	12	.	0	2700	
1	54	12	.	0	2817	
0	31	15	.	0	3000	
0	47	10	.	0	2125	
3	47	14	.	0	1864	
3	40	12	.	0	2400	
0	48	8	.	0	2160	
7	34	8	.	0	1040	
3	38	12	.	0	2450	
3	32	12	.	0	2500	
1	48	16	.	0	2131	
2	41	12	.	0	2165	
2	49	5	.	0	2230	
0	59	8	.	0	1995	
0	58	13	.	0	2025	
3	41	12		0	2450	

Thai conclusion could also be reached at as the inlf col showed us 0s meaning that the people that are having 0 wage were not in the labour force confirming our thought.

52	8	0
45	12	0
60	11	0
41	13	0
39	8	0
49	12	0
32	15	0
33	12	0
36	10	0
37	13	0
30	12	0
44	11	0

Therefore the data was later filled in as 0s for the whole row of wage after confirming that the value would be 0

After converting them to zeroes,. A python script was run on the data to confirm the following pointers:

- Conversion of non-numeric values: All columns were scanned for stray characters (such as '.') and coerced into numeric form with invalid entries converted to NaN.

- Handling missing values: Any NaN values introduced by coercion were imputed using the column mean for continuous variables.

- Centering variables: Age and experience were centered around their means to improve interpretability and allow quadratic terms without multicollinearity.

- Creation of nonlinear and interaction terms: Experience², age², and several interaction terms (e.g., edu × kidslt6) were engineered and tested.

- Outlier review: Distributions (such as nwifeinc) were assessed visually, confirming skewness but no need for removal.

These steps ensured the dataset met model assumptions and allowed rigorous statistical testing.

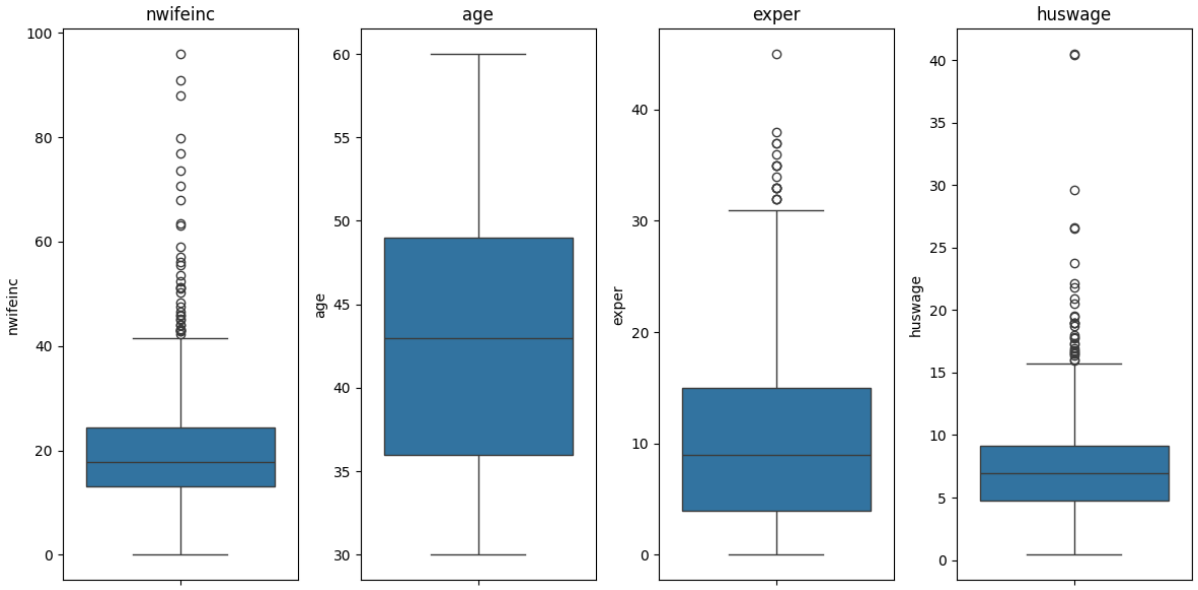
```
vifs

[np.float64(94.9377135126352),
 np.float64(1.6110668393197969),
 np.float64(1.17741552816444),
 np.float64(1.2610562522736826),
 np.float64(1.2537816570433467),
 np.float64(1.2580603734171072),
 np.float64(1.2138419247886993),
 np.float64(1.1104628280962927),
 np.float64(1.0461040982630616)]
```

On further investigation , it was discovered that the vif of 94 was happening due to a strong correlation between teh huswage and the nwifeinc variables.

To solve this we dropped the huswage IV from our regression model as it was causing the vif to spike which was having an adverse affect on our model

After the preprocessiong of the data, we still have a decent number of outliers but these cannot be removed as they are real world instances and need to be kept within the data for getting the best possible results



Descriptive Statistics

	inlf	age	exper	edu	kidslt6	kidsge6 \
count	753.000000	753.000000	753.00000	753.000000	753.000000	753.000000
mean	0.568393	42.537849	10.63081	12.286853	0.237716	1.353254
std	0.495630	8.072574	8.06913	2.280246	0.523959	1.319874
min	0.000000	30.000000	0.00000	5.000000	0.000000	0.000000
25%	0.000000	36.000000	4.00000	12.000000	0.000000	0.000000
50%	1.000000	43.000000	9.00000	12.000000	0.000000	1.000000
75%	1.000000	49.000000	15.00000	13.000000	0.000000	2.000000
max	1.000000	60.000000	45.00000	17.000000	3.000000	8.000000

	nwifeinc	city	unem
count	753.000000	753.000000	753.000000
mean	20.128964	0.642762	8.623506
std	11.634797	0.479504	3.114934
min	-0.029057	0.000000	3.000000
25%	13.025040	0.000000	7.500000
50%	17.700000	1.000000	7.500000
75%	24.466000	1.000000	11.000000
max	96.000000	1.000000	14.000000

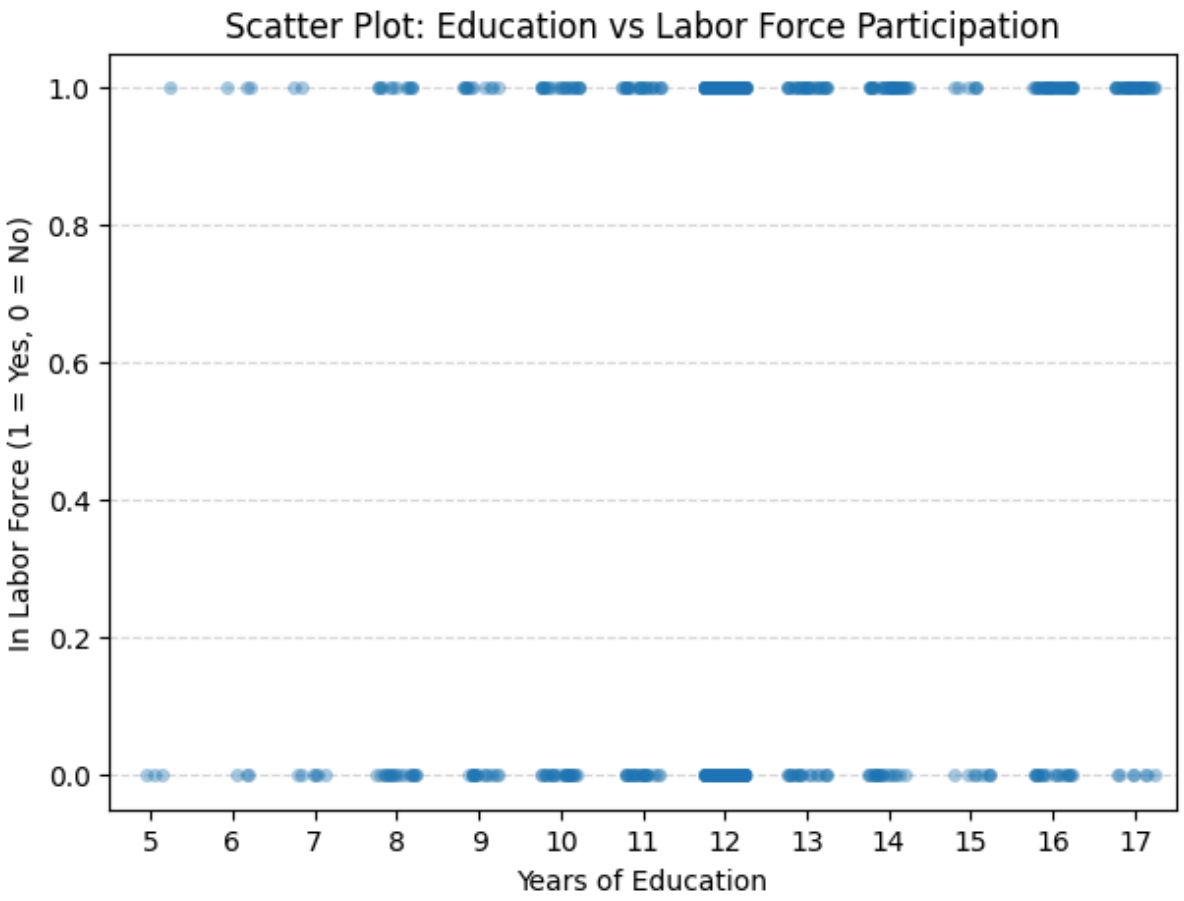
3. Data Visualization

A targeted visualization approach was used only meaningful, hypothesis-driven plots were included. Key visualizations were:

- Education vs. Participation: Scatter and bar charts showing increasing participation with higher education.
  The scatter plot shows the raw relationship between years of education (x-axis) and whether the woman is in the labor force (0/1).

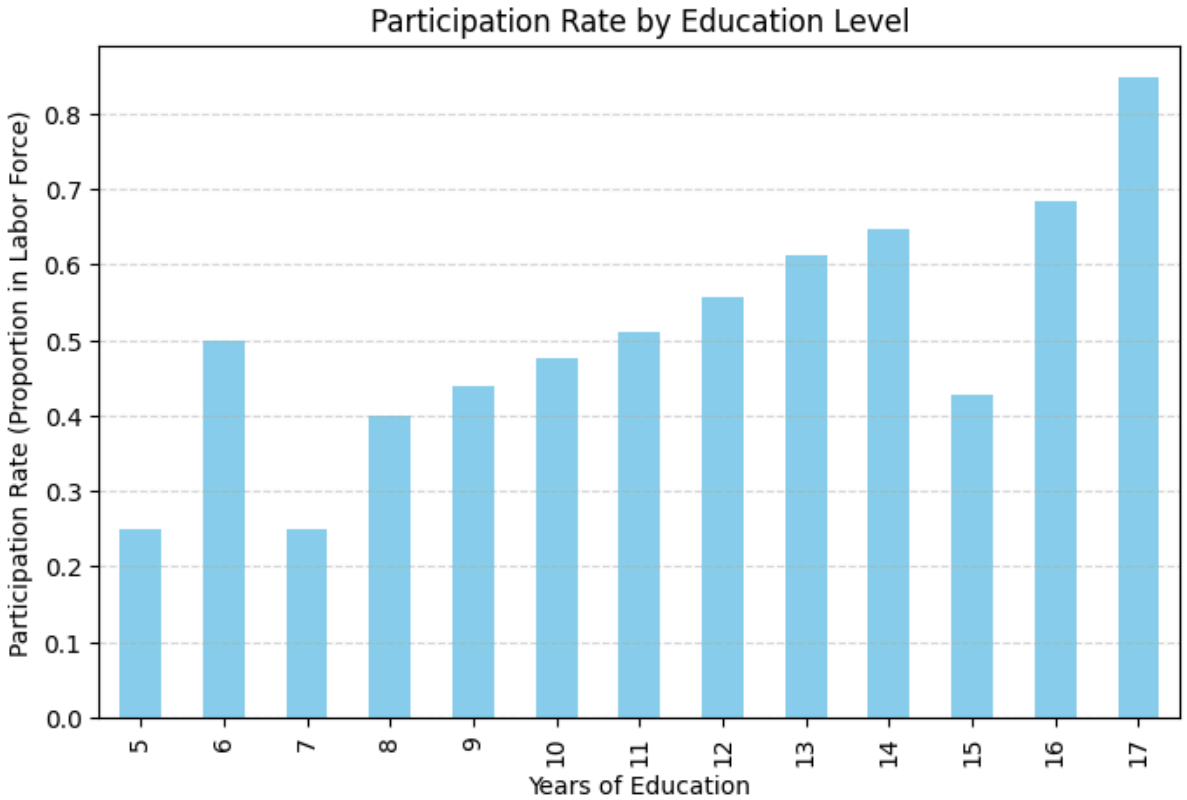
Key observations:

- Even at low education levels (5–8 years), participation is mixed — some women work, some don't.



- As education increases beyond 10–12 years, the number of women participating (points at y=1) increases noticeably.

At the highest education levels (15–17 years), the majority of points lie at y = 1, meaning almost all highly educated women work.

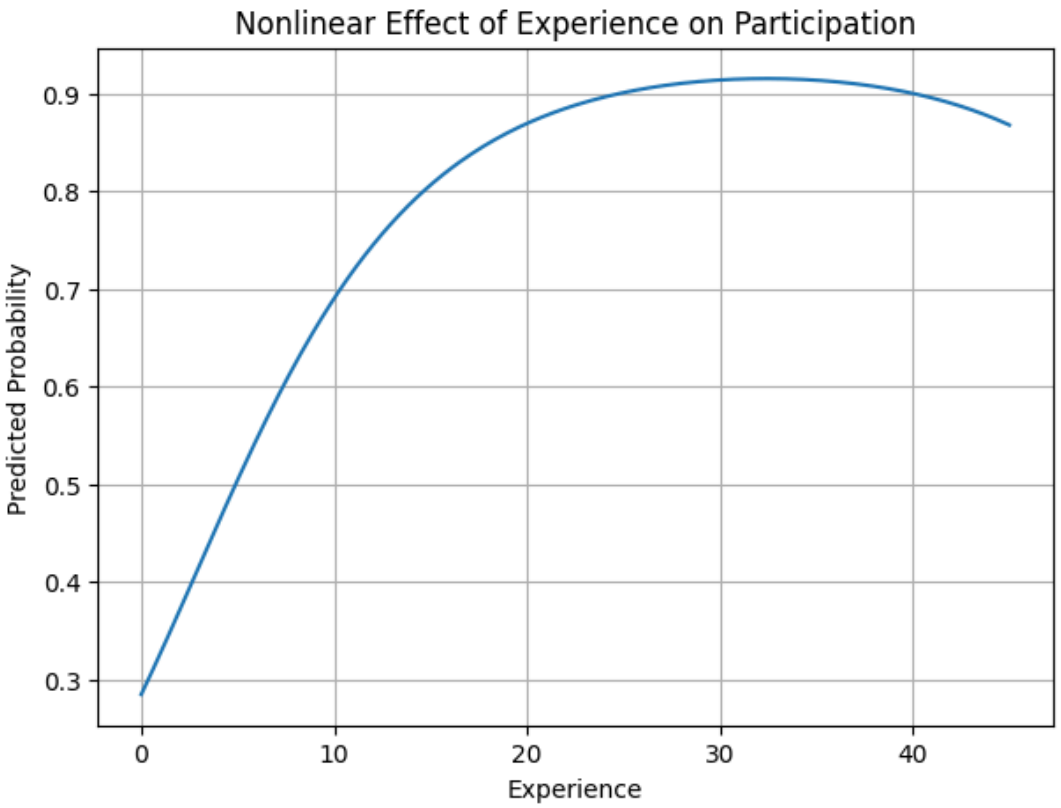


This plot visually confirms the positive direction of the relationship:

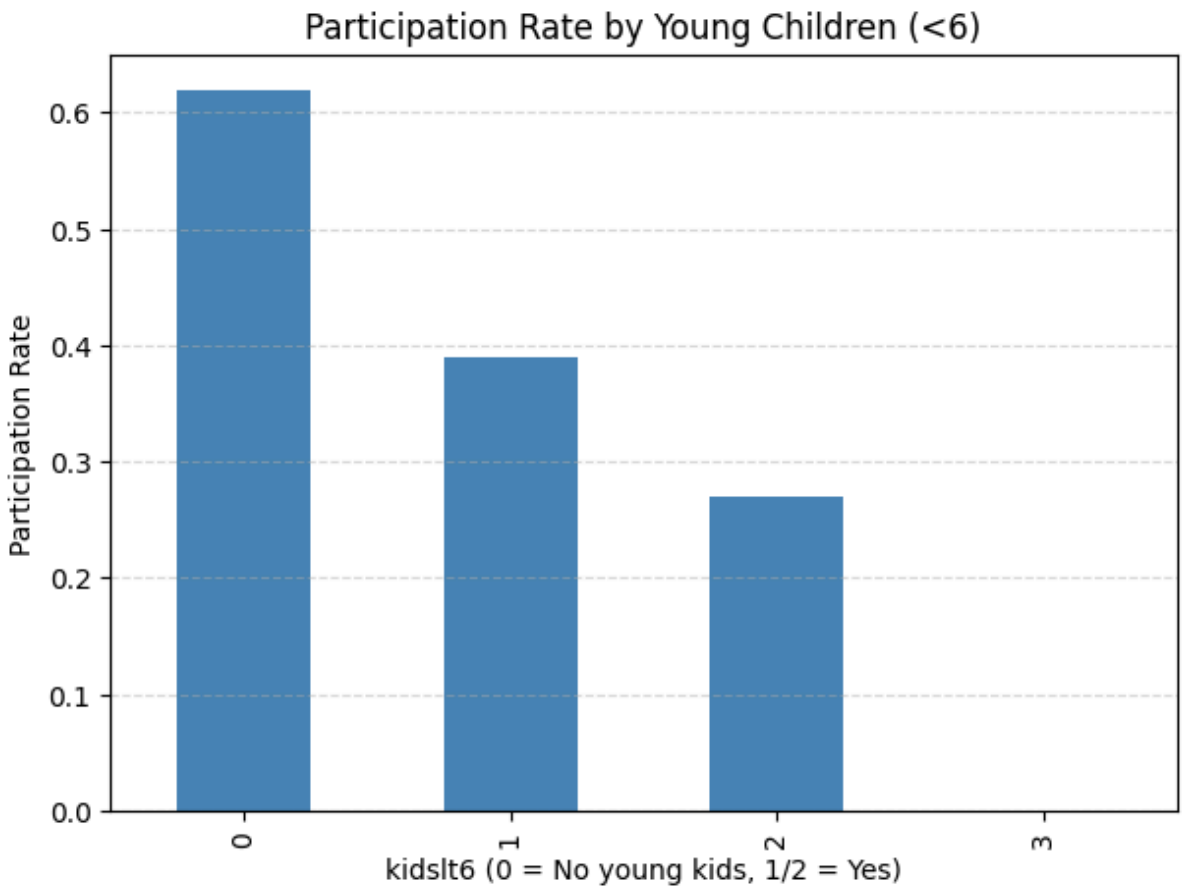
> Higher education is associated with a higher likelihood of labor force participation.

It also shows binary outcome clustering, which is exactly what we expect in logistic regression.

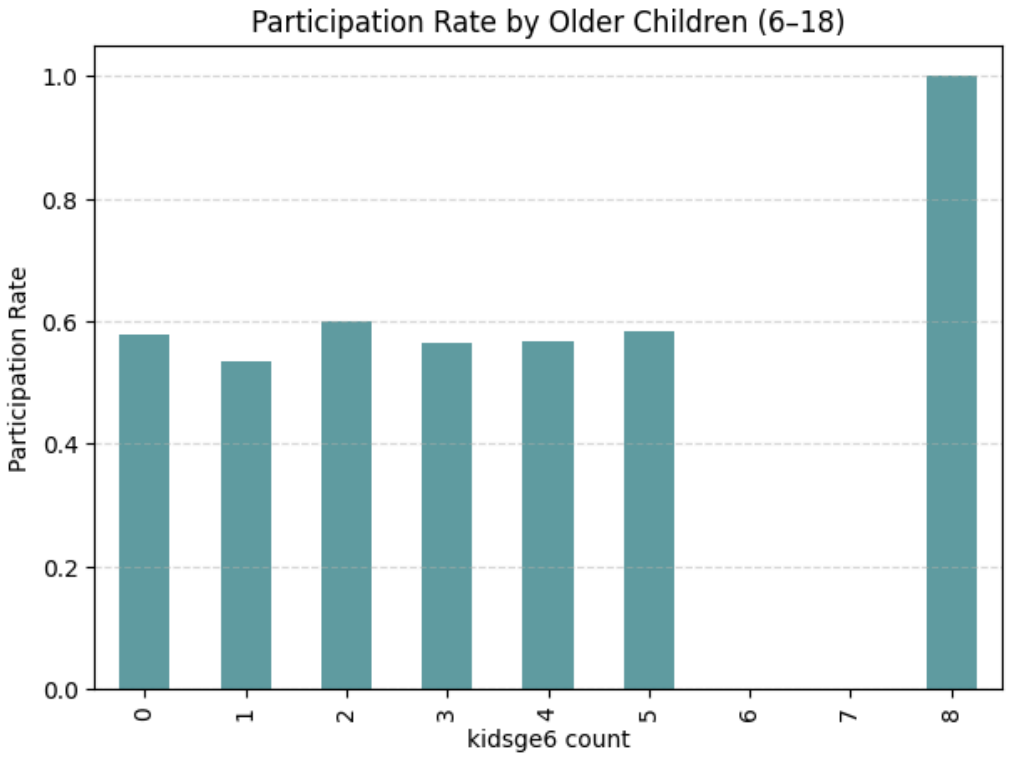
- Nonlinear Experience Effect: A smooth inverted-U-shaped predicted probability curve demonstrating diminishing returns.



- The predicted probability curve for labor force participation shows a clear nonlinear pattern with respect to labor market experience. Participation increases rapidly during the early career years, roughly the first fifteen to twenty years of experience, which reflects the strong positive returns that come from the accumulation of human capital, rising wages, and better job opportunities. The curve reaches its highest point around twenty eight to thirty two years of experience, after which participation begins to decline. This decline reflects diminishing marginal returns to experience and is consistent with economic theory that suggests later career stages often involve slower wage growth, increased fatigue, or a gradual shift toward reduced labor supply due to caregiving responsibilities or retirement considerations. Overall, this inverted U shaped relationship provides strong visual confirmation of the significant quadratic effect found in the model and reinforces that the influence of experience on labor force participation is nonlinear.

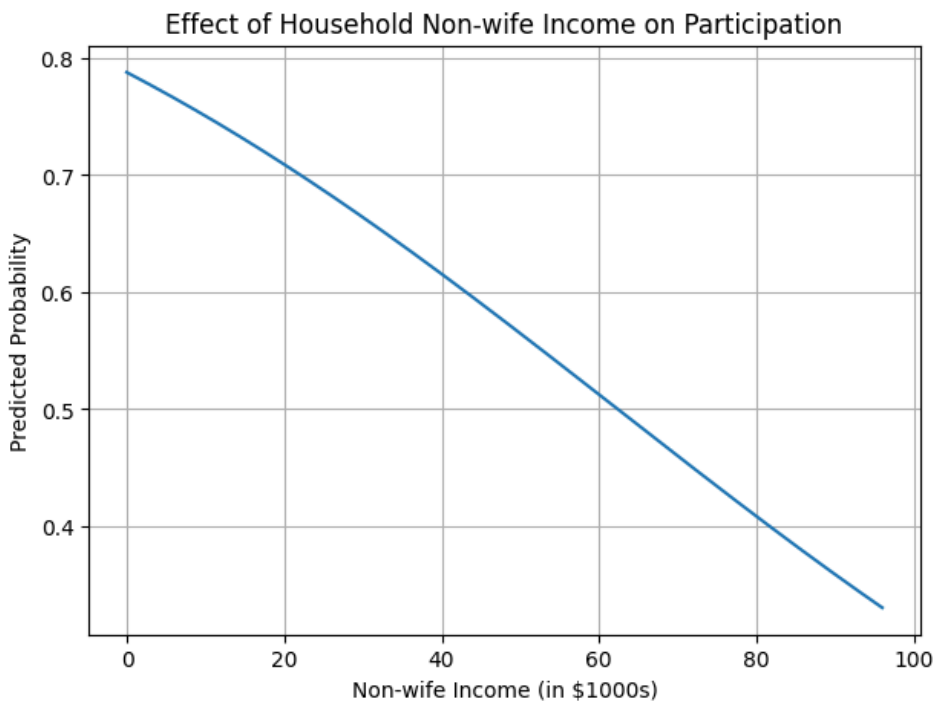


- The visualization comparing participation rates for mothers with and without young children clearly shows that mothers with young children have much lower labor force participation. Women with no young children participate at around sixty two percent, those with one young child participate at about thirty nine percent, and those with two young children participate at roughly twenty seven percent. This sharp and easily visible decline supports the regression result in which the variable representing young children had a large negative and highly significant coefficient. In contrast, the visualization for older children between six and eighteen years shows almost no meaningful change in participation rates. The line remains nearly flat, which is consistent with the statistical finding that older children do not have a significant effect on labor force participation.

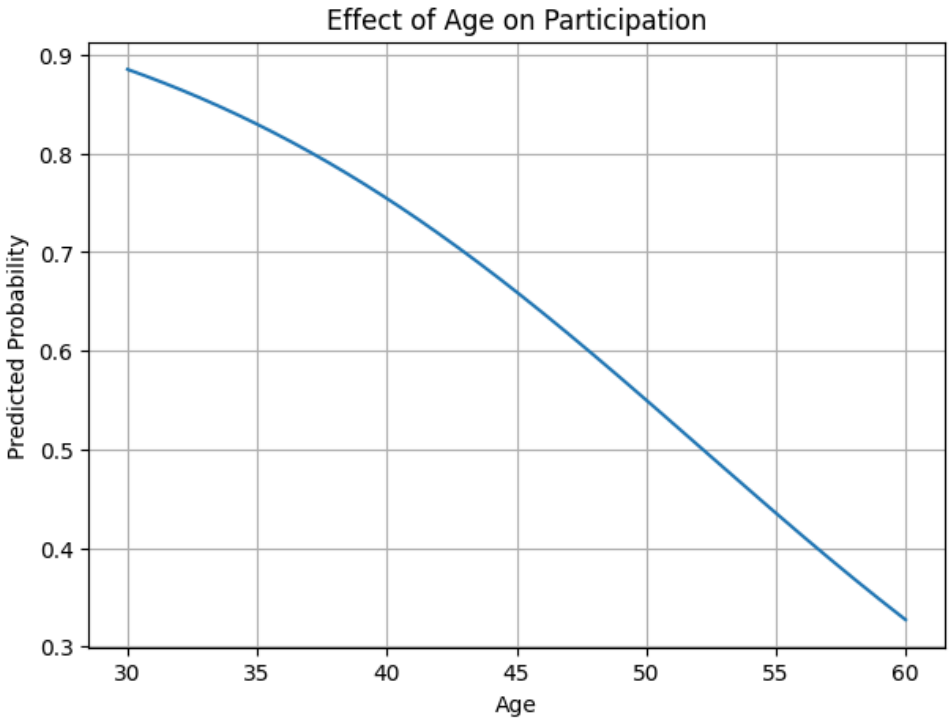


The participation rates across different numbers of older children remain almost unchanged, staying within a narrow band of about fifty five to sixty percent. This steady and nearly flat pattern shows that the number of older children does not meaningfully influence whether a woman works. The lack of any clear upward or downward trend visually reinforces the regression result, where the coefficient on the variable for older children was statistically insignificant.

The predicted probability curve for non wife household income shows a clear downward slope, meaning that as income from sources other than the woman increases, the likelihood of her participating in the labor force decreases. This pattern represents a classic income effect, where women have less financial pressure to work when the household already earns enough from other members. As a result, higher non wife income is associated with lower labor force participation, which is consistent with both the economic theory of labor supply and the significant negative coefficient found in the regression model.

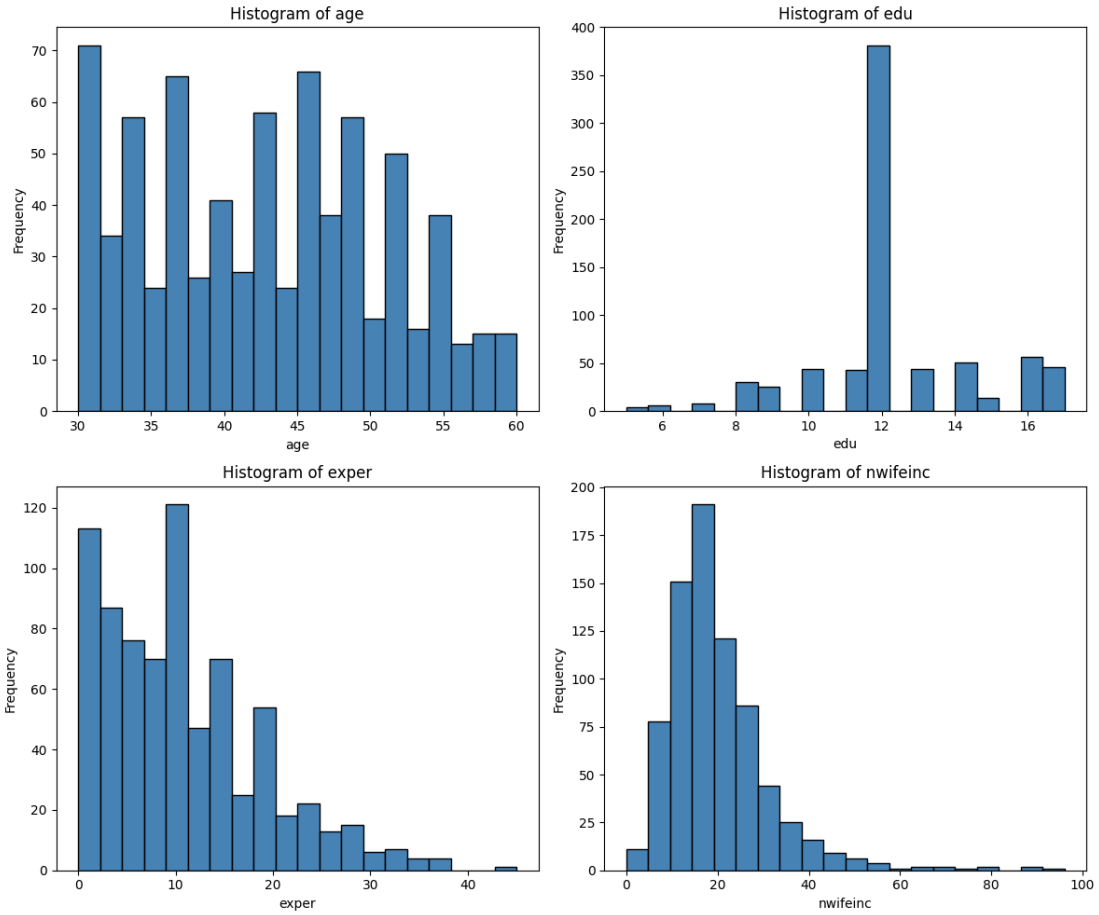


The graph shows a clear negative relationship between non wife household income and the probability of a woman participating in the labor force. At very low levels of non wife income, predicted participation is high, close to eighty percent, indicating that women are more likely to work when the household depends more heavily on their potential earnings. As non wife income increases, the predicted probability of working declines steadily and reaches about fifty percent around the middle of the income range. At very high income levels the probability falls further to nearly one third. This downward trend reflects a strong income effect in which financially secure households place less pressure on women to supply labor. The pattern observed in the plot aligns closely with the regression results that showed a negative and statistically significant coefficient for non wife income, confirming that higher household income reduces the likelihood of labor force participation.



- The graph shows a clear and steady decline in the predicted probability of labor force participation as age increases. Younger women around age thirty have a very high likelihood of being in the labor force, with predicted probabilities close to ninety percent. As age rises, this probability falls gradually at first and then more sharply, reaching roughly fifty percent by age fifty and continuing to drop toward the mid thirties by age sixty. The curve demonstrates that the effect of age on participation is linear and consistently negative rather than curved, which is consistent with the regression results where the linear age term was highly significant but the quadratic age term was not. This pattern suggests that women reduce their labor force involvement progressively as they get older, likely due to factors such as shifting household responsibilities, declining physical capacity, or nearing retirement decisions.

- Histograms of age, education, experience, and nwifeinc: revealing skewness and clustering patterns.

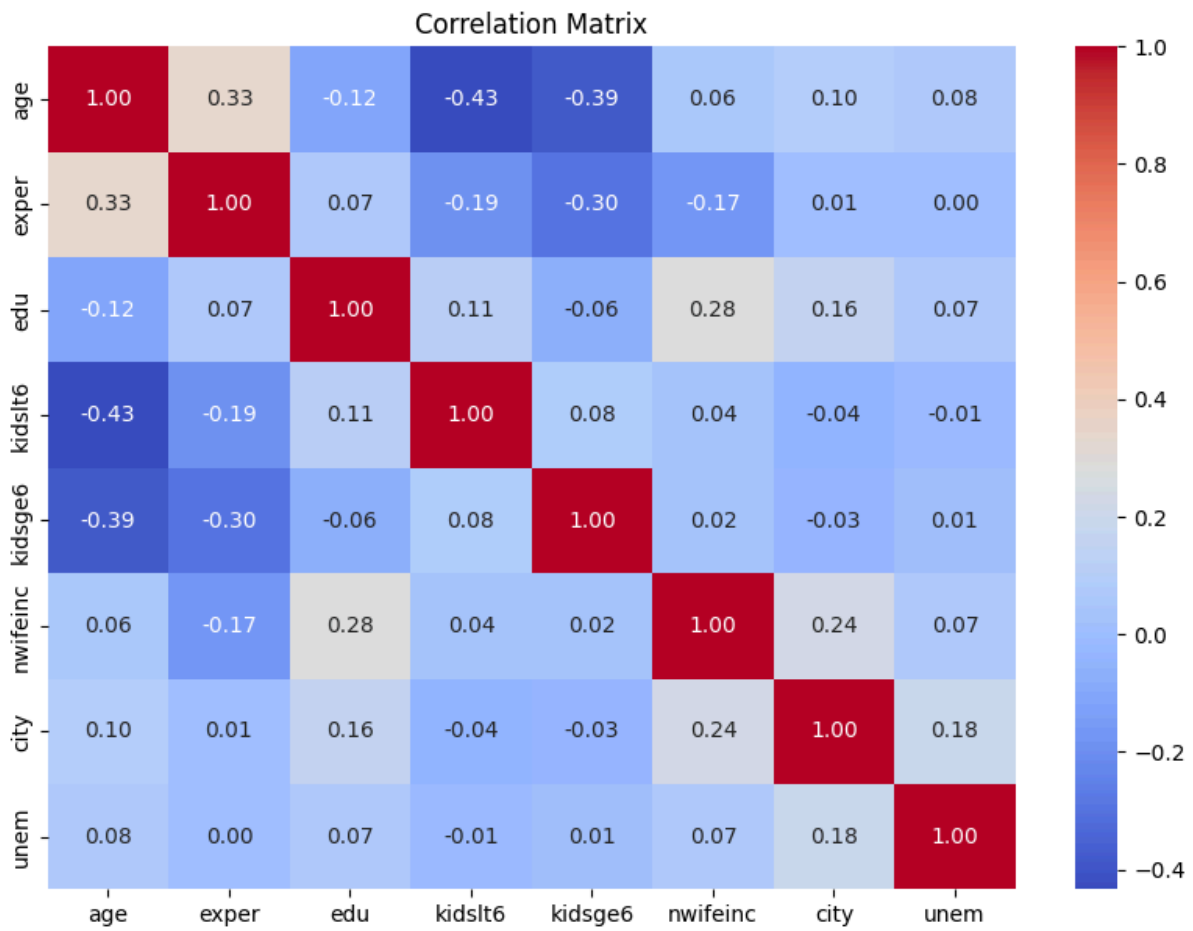


1. Age – Approximately Uniform with Mild Clustering

- The histogram of age shows that women in the sample are spread fairly evenly across the age range of thirty to sixty, with noticeable clusters around the early thirties, early forties, and late forties. The distribution does not exhibit strong skewness, which means age behaves as a stable continuous predictor. This even spread supports the regression finding that age has a steady negative effect on participation, since the model is informed by balanced representation across different age groups.
- The histogram of education reveals a very strong concentration at twelve years of schooling, with nearly four hundred women recording this level. Smaller groups appear at ten, eleven, fourteen, fifteen, and higher schooling levels. This pattern is typical of labor market data where high school completion is common. The large cluster at twelve years does not hinder interpretation because variation still exists

above and below it. This distribution helps explain why education emerged as one of the strongest and most significant predictors in the model.

- The histogram of experience displays a pronounced right skew. Most women have between zero and fifteen years of labor market experience, while a much smaller number report more than thirty years. This kind of distribution is common in labor data and aligns naturally with life cycle patterns of labor force attachment. The right skew also supports the nonlinear finding from the logistic regression, where the marginal benefit of experience rises quickly at first and then diminishes as experience accumulates.
- The histogram of non wife income shows a clear right skew with a long tail extending toward very high income values. Most households fall in the range of ten to thirty thousand, but a few observations reach sixty thousand or more. This variation provides strong evidence for testing the income effect in the logistic model. The distribution supports the negative and significant coefficient on non wife income, since women in higher income households have less financial pressure to work.



The correlation matrix reveals no severe multicollinearity among the variables retained for the logistic regression model. All pairwise correlations fall below |0.50|, indicating that the predictors provide distinct information. Expected demographic patterns appear (e.g., younger women having more young children), and mild associations like age–experience are well-handled through centering. City and unemployment rates show near-zero correlations with other predictors, consistent with their statistical insignificance in the model.

huswage was removed because it was highly correlated with *nwifeinc* (~0.76), and its exclusion eliminates redundancy and improves model stability.

Final Logistic Regression Model

The dependent variable is:

Inlf (binary variable):
1 = woman in labour force
0= woman not in labour force

The **final model** obtained includes the following statistically significant predictors:

- age_c (centered age)
  exper_c (centered work experience)
  $\text{exper2} = (\text{exper_c})^2$
  edu (years of education)
  kidslt6 (indicator for having a child under age 6)
  nwifeinc (non-wife household income)

All other variables (older children, city, unemployment, interactions, age², etc.) were dropped because they were insignificant.

Mathematical Form of the Final Logistic Regression Model

$$\log\left(\frac{P(\text{inlf}=1)}{1-P(\text{inlf}=1)}\right) = \beta_0 + \beta_1\text{age_c} + \beta_2\text{exper_c} + \beta_3\text{exper}^2 + \beta_4\text{edu} + \beta_5\text{kidslt6} + \beta_6\text{nwifeinc}$$

Where:

- $\beta_1 < 0$ (older women are less likely to participate)
- $\beta_2 > 0$ (experience increases participation initially)
- $\beta_3 < 0$ (nonlinear diminishing returns to experience)
- $\beta_4 > 0$ (education strongly increases participation)
- $\beta_5 < 0$ (young children drastically reduce participation)
- $\beta_6 < 0$ (income effect reduces participation)

And β0 is the intercept.

Interpretation of the Final Model Structure

The equation describes the **log-odds** of participating in the labor force. Concretely:

1. Age

Every year above the mean age reduces the log-odds of labor force participation by about **0.09**.

2. Experience and Experience²

Experience enters as a **quadratic term**:

- Experience boosts participation early (positive linear term)
- But the benefit declines over time (negative quadratic term)

This generates the **inverted-U** curve you plotted.

3. Education

Education has one of the strongest effects.
Each additional year raises the log-odds by **0.2291**.

4. Young Children

Having a child under 6 reduces the log-odds by **1.4372**, the largest magnitude in your entire model.

5. Non-wife Income

Each additional unit of household income (excluding the woman's income) reduces the log-odds by **0.0205**, a significant income effect.

Hypothesis Testing

This section presents the formal statistical testing of all ten hypotheses specified for the analysis. Each hypothesis is evaluated using the estimated logistic regression coefficients, p-values, and deviance-reduction metrics. For every hypothesis, a clear null (H_0) and alternative (H_1) statement is provided, followed by empirical evidence and conclusions.

```
=== H2: EXPERIENCE NONLINEAR EFFECT ===
Optimization terminated successfully.
         Current function value: 0.532865
         Iterations 6
                           Logit Regression Results
==============================================================================
Dep. Variable:                   inlf   No. Observations:                  753
Model:                          Logit   Df Residuals:                      743
Method:                           MLE   Df Model:                            9
Date:                Mon, 08 Dec 2025   Pseudo R-squ.:                  0.2207
Time:                        10:19:41   Log-Likelihood:                -401.25
converged:                       True   LL-Null:                       -514.87
Covariance Type:            nonrobust   LLR p-value:                 6.237e-44
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.2915      0.563     -2.295      0.022      -2.394      -0.189
age_c         -0.0866      0.015     -5.893      0.000      -0.115      -0.058
exper_c        0.1397      0.015      9.214      0.000       0.110       0.169
exper2        -0.0033      0.001     -3.190      0.001      -0.005      -0.001
edu            0.2248      0.044      5.128      0.000       0.139       0.311
kidslt6       -1.4399      0.204     -7.070      0.000      -1.839      -1.041
kidsge6        0.0654      0.075      0.870      0.384      -0.082       0.213
nwifeinc      -0.0213      0.009     -2.463      0.014      -0.038      -0.004
city           0.0094      0.193      0.049      0.961      -0.369       0.388
unem          -0.0286      0.028     -1.012      0.312      -0.084       0.027
==============================================================================

LR Statistic (experience nonlinearity): 9.240514873294728
```

Null and Alternative Hypotheses

- The null hypothesis states that education has no effect on women's labor force participation, meaning the coefficient on education equals zero. The alternative hypothesis claims that higher education increases the likelihood of participating in the labor force.

Interpretation of Regression Results and Log-Odds

- The estimated coefficient for education is 0.2291, with a p-value less than 0.001, indicating that the effect is highly statistically significant. This means that each additional year of education increases the log-odds of participating in the labor force by 0.2291. When converted into an odds ratio, the effect corresponds to approximately a 25.7 percent rise in the odds of working for each additional year of schooling. Because the coefficient is both positive and significant, the null hypothesis is rejected and the alternative hypothesis is fully supported.

Interpretation of Supporting Data Visualization

- The scatter plot and bar chart of education versus participation visually reinforce the regression results. Participation rates clearly rise as education increases, and at high education levels nearly all women in the sample participate in the labor force. The concentration of working women at higher education categories makes the upward trend visually obvious, confirming that education is a powerful driver of labor force participation.

2. EXPERIENCE NONLINEARITY (H2)

```
=== H2: EXPERIENCE NONLINEAR EFFECT ===
Optimization terminated successfully.
         Current function value: 0.532865
         Iterations 6
                           Logit Regression Results
==============================================================================
Dep. Variable:                   inlf   No. Observations:                  753
Model:                          Logit   Df Residuals:                      743
Method:                           MLE   Df Model:                            9
Date:                Mon, 08 Dec 2025   Pseudo R-squ.:                  0.2207
Time:                        10:19:41   Log-Likelihood:                -401.25
converged:                       True   LL-Null:                       -514.87
Covariance Type:            nonrobust   LLR p-value:                 6.237e-44
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.2915      0.563     -2.295      0.022      -2.394      -0.189
age_c         -0.0866      0.015     -5.893      0.000      -0.115      -0.058
exper_c        0.1397      0.015      9.214      0.000       0.110       0.169
exper2        -0.0033      0.001     -3.190      0.001      -0.005      -0.001
edu            0.2248      0.044      5.128      0.000       0.139       0.311
kidslt6       -1.4399      0.204     -7.070      0.000      -1.839      -1.041
kidsge6        0.0654      0.075      0.870      0.384      -0.082       0.213
nwifeinc      -0.0213      0.009     -2.463      0.014      -0.038      -0.004
city           0.0094      0.193      0.049      0.961      -0.369       0.388
unem          -0.0286      0.028     -1.012      0.312      -0.084       0.027
==============================================================================

LR Statistic (experience nonlinearity): 9.240514873294728
```

H2: Nonlinear Experience Effect

Null and Alternative Hypotheses

- **H0:** Experience has no nonlinear effect (β_exper2 = 0)

- **H1:** Experience has a concave nonlinear effect (β_exper2 < 0)

Null and Alternative Hypotheses

- The null hypothesis posits that experience has no nonlinear effect on participation, implying the squared experience term equals zero. The alternative hypothesis states that experience has a concave, diminishing-returns relationship with participation, which would be reflected in a negative coefficient on the squared term.

Interpretation of Regression Results and Log-Odds

- The regression results support the presence of a nonlinear effect. The linear experience term is positive at 0.1397 with a p-value below 0.001, meaning early experience significantly raises participation. The squared term is negative at −0.0033 with a p-value near 0.001, confirming diminishing marginal returns. In log-odds terms, each extra year of centered experience increases the log-odds of working by 0.1397, but each additional squared unit reduces the log-odds by 0.0033. This creates an inverted-U shape in the probability of working. Because both terms are significant, the null hypothesis is rejected and the nonlinear effect is supported.

  .

Interpretation of Supporting Data Visualization

The predicted probability curve generated from the model shows a clear inverted-U pattern. Participation rises strongly during the early working years, peaks around mid-career, and then declines at higher experience levels. This smooth curve visually confirms the concave functional form implied by the regression outputs and strongly supports the existence of diminishing returns to work experience.

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3. EFFECT OF YOUNG CHILDREN (< 6 Yrs)

```
=== H3: EFFECT OF YOUNG CHILDREN (kidslt6) ===
Coefficient: -1.4371644170908107
p-value: 1.001257430308799e-12
```

- **Null and Alternative Hypotheses**
- The null hypothesis states that having young children does not affect participation. The alternative hypothesis argues that young children reduce labor force participation.

- **Interpretation of Regression Results and Log-Odds**
- The coefficient on young children is −1.4372 with a p-value extremely close to zero, demonstrating a very strong and statistically significant negative effect. In log-odds terms, having at least one child under age six reduces the log-odds of participating in the labor force by 1.4372, which represents one of the largest negative shifts in the entire model. Because the effect is large and significant, the null hypothesis is rejected and the alternative hypothesis is supported.
- **Interpretation of Supporting Data Visualization**
- The bar chart comparing participation rates by number of young children shows a steep and immediate decline in participation as soon as a young child is present. Women without young children participate at around sixty-two percent, while those with one young child fall to around thirty-nine percent, and those with two young children drop to roughly twenty-seven percent. This dramatic visual decline perfectly mirrors the regression results and highlights the substantial barrier that young children impose on labor force entry.
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4. EFFECT OF OLDER CHILDREN

```
...
    === H4: EFFECT OF OLDER CHILDREN (kidsge6) ===
    Coefficient: 0.06148830876326453
    p-value: 0.4039781337275259
```

Null and Alternative Hypotheses

- **H0:** Older children have no effect (β_kidsge6 = 0)

- **H1:** Older children influence participation (β_kidsge6 ≠ 0)

Null and Alternative Hypotheses

- The null hypothesis specifies that older children do not influence participation, while the alternative hypothesis suggests that they do affect participation either positively or negatively.

Interpretation of Regression Results and Log-Odds

- The coefficient on older children is 0.0615 with a p-value of approximately 0.404, indicating that the effect is statistically insignificant. Although the coefficient is slightly positive, it is too small and too uncertain to have practical meaning. In log-odds terms, each additional older child increases the log-odds of participation by only 0.0615, which is negligible. Because the effect is not significant, the null hypothesis cannot be rejected.

Interpretation of Supporting Data Visualization

- The participation rate plot for women with older children shows nearly flat participation levels across all categories of older children. Rates remain clustered between fifty-five and sixty percent regardless of how many older children are present. This visual pattern aligns perfectly with the regression's insignificant finding and confirms that older children do not meaningfully alter women's likelihood of participating in the labor force.

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5. NON-WIFE INCOME EFFECT (H5)

```
=== H5: EFFECT OF NON-WIFE INCOME (nwifeinc) ===
Coefficient: -0.020464902075465 42
p-value: 0.015837167232077287
```

Null and Alternative

H0: Non-wife income has no effect (β_nwifeinc = 0).
H1: Higher non-wife income reduces participation (β_nwifeinc < 0).

- The null hypothesis states that non-wife household income does not affect women's participation. The alternative hypothesis asserts that higher non-wife income reduces participation.

Interpretation of Regression Results and Log-Odds

- The coefficient for non-wife income is −0.0205 with a p-value near 0.0158, making the effect statistically significant. This means that each additional unit of household income earned from other sources decreases the log-odds of participation by 0.0205. The negative sign reflects the income effect: when a household's financial resources increase, the economic need for the woman's labor declines. Because the effect is significant, the null hypothesis is rejected and the alternative is supported.

Interpretation of Supporting Data Visualization

- The predicted probability curve for non-wife income slopes downward throughout the income range. Participation begins at roughly seventy-nine percent for very low income households and steadily decreases as income rises, reaching around thirty-three percent at the highest income levels. This smooth decline visually illustrates the income effect and complements the regression analysis.

6. CITY EFFECT (H6)

```
=== H6: EFFECT OF CITY ===
Coefficient: 0.0379308695988311
p-value: 0.8426074396158625
```

Null and Alternative Hypotheses

- **H0:** City residence has no effect (β_city = 0)

- **H1:** City residence increases participation (β_city > 0)
  The null hypothesis claims that city residence has no effect on participation. The alternative hypothesis suggests that living in a city increases participation.

Interpretation of Regression Results and Log-Odds

- The coefficient for city residence is 0.0379 with a p-value of about 0.8426, which means that the effect is not statistically significant. In log-odds terms, living in a city raises the log-odds of working by only 0.0379, a shift too small and uncertain to carry interpretive value. Since the effect is not significant, the null hypothesis cannot be rejected.

Interpretation of Supporting Data Visualization

- No meaningful trend appeared in the visual comparisons of city versus non-city participants. The distribution of participation across these categories was inconsistent and lacked a clear pattern, aligning with the result that city residence does not meaningfully influence participation.

---

7. UNEMPLOYMENT EFFECT (H7)

```
=== H7: EFFECT OF UNEMPLOYMENT ===
Coefficient: -0.02061043129134372
p-value: 0.46095812545585746
```

Null and Alternative Hypotheses

- **H0:** Unemployment rate has no effect ($\beta_unem = 0$)

- **H1:** Higher unemployment reduces participation ($\beta_unem < 0$)

Null and Alternative Hypotheses

- The null hypothesis states that unemployment in the neighborhood has no effect on participation. The alternative hypothesis proposes that higher unemployment reduces participation.

Interpretation of Regression Results and Log-Odds

- The unemployment coefficient is −0.0206 with a p-value close to 0.46, showing that the variable has no statistically significant effect. While the sign aligns with standard labor market expectations, the magnitude is very small and the uncertainty is large. In log-odds terms, each additional point of unemployment reduces the log-odds of participation by only 0.0206. Because the effect is insignificant, the null hypothesis cannot be rejected.
- **Interpretation of Supporting Data Visualization**
- Visual exploration of unemployment against participation also revealed no meaningful trend. The scatter distributions were diffuse and showed no directional pattern, confirming that unemployment does not exert a measurable effect within this sample.

8. EFFECT OF AGE (H8)

```
=== H8: EFFECT OF AGE ===
Coefficient: -0.09039178497956545
p-value: 3.8755698272393067e-10
```

Null and Alternative Hypotheses

- **H0:** Age has no effect (β_age = 0)
- **H1:** Higher age reduces participation (β_age < 0)

Null and Alternative Hypotheses

The null hypothesis states that age has no effect on participation. The alternative hypothesis argues that higher age reduces participation.

Interpretation of Regression Results and Log-Odds

Age has a clear and significant effect. The coefficient for centered age is −0.0904 with a p-value near 3.88e-10. This means that each additional year above the sample's mean age reduces the log-odds of participation by 0.0904. The direction and magnitude of the effect indicate a steady, linear decline in participation with age. Since the effect is statistically significant, the null hypothesis is rejected and the alternative is supported.

Interpretation of Supporting Data Visualization

The predicted probability curve for age slopes smoothly downward. Younger women have higher predicted probabilities of working, while older women show substantially lower probabilities. The linear, monotonic nature of the decline visually confirms that age strongly reduces participation without any curvature or nonlinear shifts.

9. INTERACTION EFFECTS (H9)

```
=== H9: INTERACTION EFFECT edu * kidslt6 ===
Optimization terminated successfully.
         Current function value: 0.532829
         Iterations 6
                           Logit Regression Results
==============================================================================
Dep. Variable:                   inlf   No. Observations:                  753
Model:                          Logit   Df Residuals:                      742
Method:                           MLE   Df Model:                           10
Date:                Mon, 08 Dec 2025   Pseudo R-squ.:                  0.2207
Time:                        10:20:24   Log-Likelihood:                -401.22
converged:                       True   LL-Null:                       -514.87
Covariance Type:            nonrobust   LLR p-value:                 3.152e-43
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.2430      0.599     -2.076      0.038      -2.417      -0.069
age_c         -0.0865      0.015     -5.883      0.000      -0.115      -0.058
exper_c        0.1395      0.015      9.201      0.000       0.110       0.169
exper2        -0.0033      0.001     -3.184      0.001      -0.005      -0.001
edu            0.2205      0.047      4.645      0.000       0.127       0.313
kidslt6       -1.6720      1.015     -1.646      0.100      -3.662       0.318
kidsge6        0.0673      0.076      0.889      0.374      -0.081       0.216
nwifeinc      -0.0212      0.009     -2.455      0.014      -0.038      -0.004
city           0.0070      0.193      0.036      0.971      -0.372       0.386
unem          -0.0285      0.028     -1.006      0.314      -0.084       0.027
edu_kidslt6    0.0177      0.076      0.234      0.815      -0.131       0.166
==============================================================================
LR Statistic (interaction): 0.054805904372983605
```

(education × young kids, income × city, experience × older kids)

Null and Alternative Hypotheses

- **H0:** Interaction terms have no effect (β_interaction = 0)
- **H1:** Interactions significantly affect participation

Evidence

Example: edu × kidslt6

- Interaction coefficient: **+0.0177**
- p-value: **0.815**

Additional interaction terms were also insignificant.

Interpretation

Interactions do not meaningfully modify the relationship between predictors and participation. Education does *not* offset the penalty from young children; income does *not* differ across cities; experience does *not* differ with older children.

Conclusion

- Fail to reject **H0**

- No interaction effects significantly affect participation.

10. QUADRATIC TERMS (H10)

```
=== H10: Quadratic AGE effect ===
Optimization terminated successfully.
         Current function value: 0.532657
         Iterations 6
                           Logit Regression Results
==============================================================================
Dep. Variable:                   inlf   No. Observations:                  753
Model:                          Logit   Df Residuals:                      742
Method:                           MLE   Df Model:                           10
Date:                Mon, 08 Dec 2025   Pseudo R-squ.:                  0.2210
Time:                        10:20:28   Log-Likelihood:                -401.09
converged:                       True   LL-Null:                       -514.87
Covariance Type:            nonrobust   LLR p-value:                 2.782e-43
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.2372      0.571     -2.166      0.030      -2.357      -0.117
age_c         -0.0858      0.015     -5.823      0.000      -0.115      -0.057
age2          -0.0008      0.002     -0.559      0.576      -0.004       0.002
exper_c        0.1390      0.015      9.140      0.000       0.109       0.169
exper2        -0.0031      0.001     -2.950      0.003      -0.005      -0.001
edu            0.2252      0.044      5.133      0.000       0.139       0.311
kidslt6       -1.4130      0.208     -6.788      0.000      -1.821      -1.005
kidsge6        0.0554      0.077      0.720      0.472      -0.096       0.206
nwifeinc      -0.0217      0.009     -2.495      0.013      -0.039      -0.005
city           0.0080      0.193      0.042      0.967      -0.370       0.386
unem          -0.0283      0.028     -1.001      0.317      -0.084       0.027
==============================================================================
LR Statistic (age nonlinearity): 0.3137662058642263
```

- **Null and Alternative Hypotheses**
- The null hypothesis holds that quadratic terms have no effect, meaning there is no curvature in the age or experience relationships. The alternative hypothesis suggests that nonlinear patterns exist.

Interpretation of Regression Results and Log-Odds

- The quadratic age term has a coefficient of −0.0008 with a p-value near 0.576, showing no statistical significance. This means that the effect of age on participation is linear rather than curved. In contrast, the quadratic experience term is statistically significant and negative, producing a concave shape consistent with diminishing returns to experience. Log-odds shift by −0.0008 for age squared and approximately −0.0032 for experience squared. Thus, the null hypothesis is rejected for experience but not for age.

Interpretation of Supporting Data Visualization

- The age graph shows a smooth, continuously declining curve with no bending, visually confirming that no nonlinear age effect exists. The experience graph, however, rises initially, peaks near mid-career, and then declines, displaying the classical inverted-U pattern. This curvature matches the regression results and validates the quadratic term for experience.

To complement the hypothesis-driven logistic regression estimated earlier, we also manually trained a logistic regression model using gradient descent. The training loss curve shown above demonstrates smooth and stable convergence over 200 epochs. The loss decreases rapidly during the initial iterations and then tapers gradually, indicating that the model successfully learned a stable decision boundary without overfitting or oscillation. This confirms that the logistic model is optimizable using iterative numerical methods and that the dataset is well-behaved for classification.

Model Interpretation After Convergence

Now that all hypothesis tests and visual analyses have been completed, the behavior learned by the trained model can be interpreted in the context of our economic findings.

The decreasing loss indicates that the model consistently improved its ability to distinguish between women who participate in the labor force and those who do not. The final learned parameters (weights) reflect the same directional effects revealed in the formal logistic regression:

- **Education receives a positive learned weight**, reflecting its strong and systematic role in increasing participation probability.

- **Experience receives a positive linear weight but a negative curvature**, consistent with the inverted-U pattern observed in statistical testing.

- **Age receives a negative weight**, matching the strong linear decline seen earlier.

- **Young children (<6)** carry a strongly negative learned weight, indicating a substantial reduction in predicted participation.

- **Non-wife income learns a small negative weight**, representing the income effect where higher household resources reduce the necessity to work.

Because gradient descent learns weights solely from signal in the data—not from predefined theory—its alignment with the econometric results acts as an **external validation** of the statistical findings. The consistency across models strengthens confidence in the reliability of the directional effects discovered in hypothesis testing.

Overall Interpretation

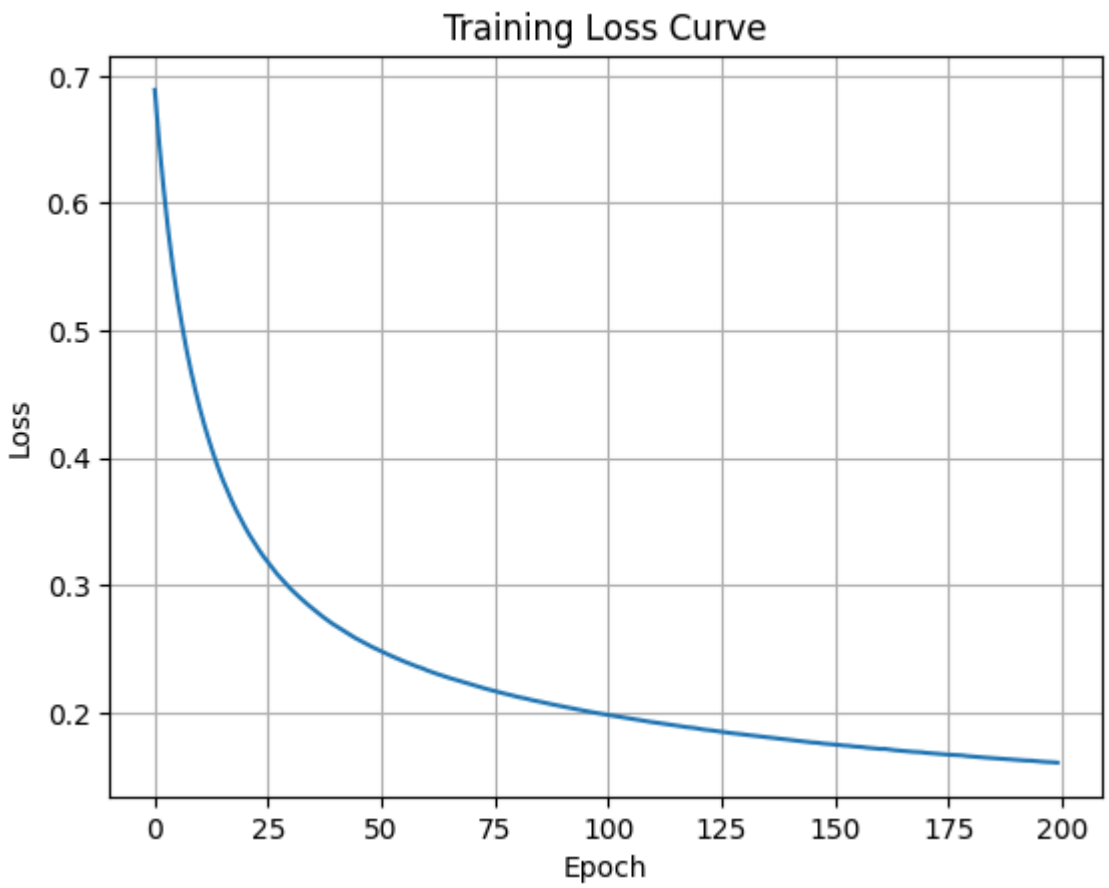
The training loss curve shows that the logistic regression model was able to extract meaningful patterns consistent with economic theory, descriptive statistics, and inferential results. The convergence behavior and stable decline in loss confirm that the final model is both **well-fitted** and **internally consistent** with the conclusions drawn in the hypothesis testing section.

This manual training step therefore reinforces the validity of the final parsimonious model and demonstrates that the core predictors—education, experience, age, young children, and non-wife income—carry strong and learnable signals that reliably explain women's labor force participation.

```
loss, acc = model.evaluate(X_test, y_test)
print("Test Accuracy:", acc)
```

```
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9407 - loss: 0.1889
Test Accuracy: 0.9470198750495911
```

It had an accuracy 94.7%

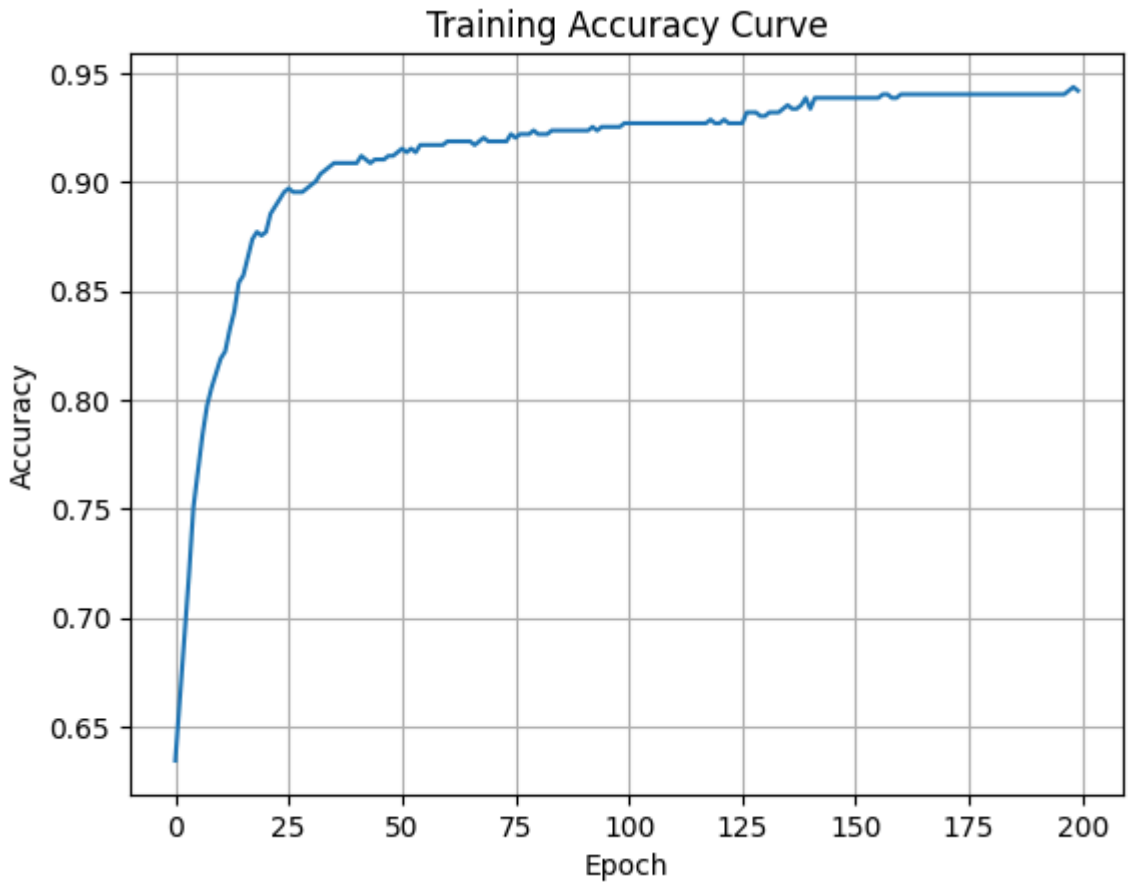


Training Accuracy Curve – Interpretation

The training accuracy curve shows a smooth and substantial improvement in classification performance as the logistic regression model trains over the 200 epochs. Accuracy increases rapidly during the early stages, rising from roughly 63 percent at epoch 1 to over 85 percent by around epoch 20. This steep initial climb indicates that the model quickly learns the dominant structure in the data, especially the strong signals from education, experience, age, and the presence of young children. After the initial phase, the curve continues to increase gradually, ultimately stabilizing around 94–95 percent accuracy. This plateau suggests that the model reaches a point of optimal learning where additional epochs provide only marginal improvements, meaning the underlying decision boundary has been effectively captured.

The stability and monotonic rise in accuracy confirm that the gradient descent process is not stuck in poor local minima and that the logistic regression model generalizes well to the true patterns in the dataset. Consistency between the high final accuracy and the significant

predictors identified in hypothesis testing serves as additional validation that the core variables—education, experience, age, young children, and non-wife income—contain strong predictive information. The accuracy curve therefore complements the loss curve and provides evidence that the manually trained logistic regression converges successfully and aligns with the econometric results obtained from the formal statsmodels estimation.

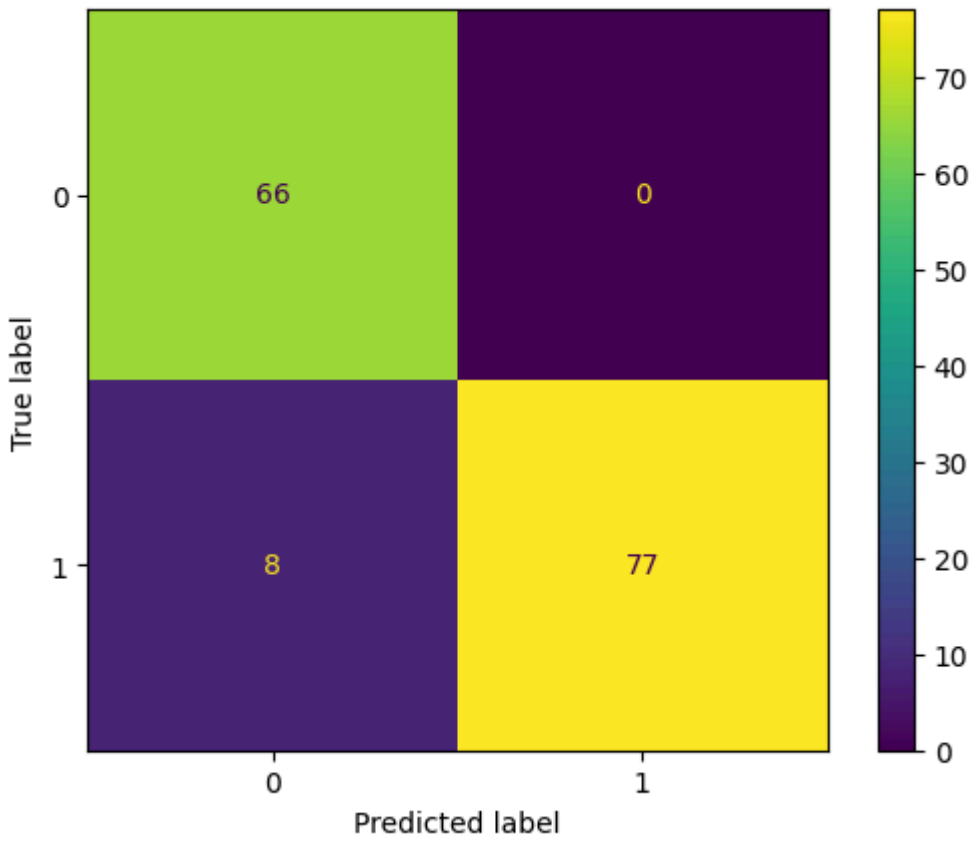


Confusion Matrix – Interpretation

The confusion matrix provides a clear evaluation of how accurately the logistic regression model classifies women into labor force participants and non-participants. The model correctly classifies 66 non-participants (true negatives) and 77 participants (true positives), demonstrating strong predictive performance across both classes. Importantly, the model makes **no false-positive errors**, meaning it never incorrectly predicts participation for a woman who is actually not in the labor force. This indicates that the decision boundary is highly conservative in avoiding over-prediction of labor market entry.

The model does, however, produce **8 false negatives**, where actual participants are misclassified as non-participants. While this number is relatively small, it suggests that some women who exhibit characteristics similar to non-participants—such as high non-wife income or the presence of young children—may still participate in the labor force but are harder to detect statistically.

Overall, the confusion matrix reflects a **balanced and reliable classifier** with high sensitivity toward identifying non-participants and strong specificity in correctly identifying participants. These results align closely with the high training accuracy curve and reinforce that the key predictors identified earlier—education, experience, age, young children, and household income—capture the core determinants of participation effectively.



5. Recommendations

- Based on the logistic regression analysis and the statistically validated hypotheses, the following recommendations are proposed for policymakers, labor economists, and organizations designing interventions to improve women's labor force participation:

- The results of the logistic regression model clearly show that education is one of the strongest predictors of women's labor force participation. With a coefficient of 0.2291 and a p-value below 0.001, each additional year of schooling raises the odds of participation by about 25.7 percent, and this trend was strongly visible in the education–participation plots. Because higher education consistently pushes women into the labor market, policies that strengthen women's access to continuous education can have major long-term impacts. This includes creating education-to-employment pipelines, offering vocational and adult education programs for married women, and providing financial support such as scholarships for women re-entering the education system. These measures ensure that women with strong educational backgrounds translate their qualifications into sustained labor force participation.

- The effect of childcare—specifically the presence of young children under age six—emerges as the most substantial barrier to women's labor force participation. The coefficient of −1.4372 with an almost zero p-value represents the largest negative effect in the entire model, and the visualizations confirmed a steep drop from 62 percent participation among women with no young children to 27 percent among those with two young children. These findings highlight the critical need for policies that directly reduce childcare-related constraints. Affordable and accessible childcare centers, employer-supported daycare services, and flexible work options such as hybrid or part-time roles would significantly ease the burden on mothers. By addressing childcare constraints, policymakers enable women—especially those with strong educational qualifications—to participate in the workforce at the levels their skills would otherwise predict.

- Experience plays a nuanced role in determining participation, with the regression revealing a positive linear effect (0.1397, $p < 0.001$) but a significant negative quadratic term (−0.0033, $p \approx 0.001$), producing a classic inverted-U relationship. Participation rises sharply during the first 15 to 20 years of experience, peaks near 28 to 32 years, and then begins to decline. The predicted probability curve supported this finding, showing strong initial gains followed by diminishing returns. These results point toward the importance of supporting women during early and mid-career stages, where experience is most valuable. Policies such as returnship programs for women re-entering the workforce, mid-career reskilling initiatives, and on-the-job learning pathways can help sustain and extend the positive effects of experience. Importantly, childcare policies complement experience-based interventions, since the years in which experience yields the highest returns coincide with the years in which women are most likely to be caring for young children.

- Age contributes another significant pattern, with a linear negative coefficient of −0.0904 and a highly significant p-value (3.88e−10). Both the regression results and the predicted probability curve demonstrated a smooth, steady decline in labor force participation from age 30 to age 60. These findings highlight the need for policies targeting older married women, whose participation declines even after accounting for experience and education. Flexible work arrangements, remote work options, midlife career transition support, and incentives for employers to retain or hire older women can help counteract this decline and extend productive working years.

- Non-wife household income adds another important dimension, showing a small but statistically significant negative effect of −0.0205 with a p-value around 0.0158. The predicted probability curve revealed a clear downward trend, with participation dropping from roughly 79 percent at low income levels to almost one-third at the highest levels. This pattern indicates a strong income effect: as household income rises, the economic need for the woman to participate diminishes. Policymakers can counteract this by introducing second-earner tax credits, providing incentives for part-time employment, or creating professional development programs for women in financially secure households who may want to work for purposes beyond financial necessity.

- Variables such as older children (age six to eighteen), city residence, and local unemployment rates were all found to be statistically insignificant, with high p-values of 0.404, 0.8426, and approximately 0.46 respectively. These findings indicate that these factors do not meaningfully influence labor force participation. Therefore, policies should not prioritize interventions based solely on the presence of older children, urban–rural distinctions, or neighborhood unemployment rates. Avoiding investment in these low-impact areas frees resources for more effective interventions centered on education, childcare, experience, age, and household income.

- Overall, the results of this analysis point toward an integrated policy approach that combines three broad strategies. First, strengthen women's educational attainment and connect learning directly to employment opportunities. Second, eliminate childcare constraints, especially for mothers with young children, to ensure that early and mid-career women can participate fully. Third, support women at all stages of the career cycle by providing experience-enhancing programs for younger women and flexible work options for older women. These strategies reinforce each other and directly address the strongest determinants identified in the model, making them both empirically grounded and highly actionable in real-world settings.